Lecture 8 2025-09-23

p is a <u>fixed point</u> of $g(x)$ if $g(p)=p$

When does g have a (unique) fixed point?

Sufficient conditions for <u>existence</u>:

1.) g is continuous on $[a,b]$

2.) g maps $[a,b]$ to $[a,b]$ $(a \le g(x) \le b$ for $x \in [a,b])$

Sufficient condition for <u>uniqueness</u>:

1, 2 and 3.) g is differentiable on (a,b) and $\exists\, k<1$ such that $|g'(x)| \le k$ for all $x \in (a,b)$ $(k \in [0,1))$

<u>Proof of existence:</u>

Let $h(x) = g(x) - x$. (h has a root $\iff$ g has a fixed point)

$\Rightarrow$ Must show that $\exists\, p \in [a,b]$ such that $h(p)=0$.

If $h(a)=0$ or $h(b)=0$, we are done (set $p=a$ or $p=b$)

Otherwise since range$(g) \subset [a,b]$, we have

$h(a) = g(a) - a > 0$

$h(b) = g(b) - b < 0$ $\Rightarrow$ opposite signs

h is continuous since g is. Therefore by IVT $\exists\, p \in (a,b)$ s.t. $h(p)=0 \iff g(p)=p$ ∎

<u>Uniqueness proof</u> (assuming 3.) also)

$h(x) = g(x) - x$. Assume by contradiction that $h(x_1)=0$ and

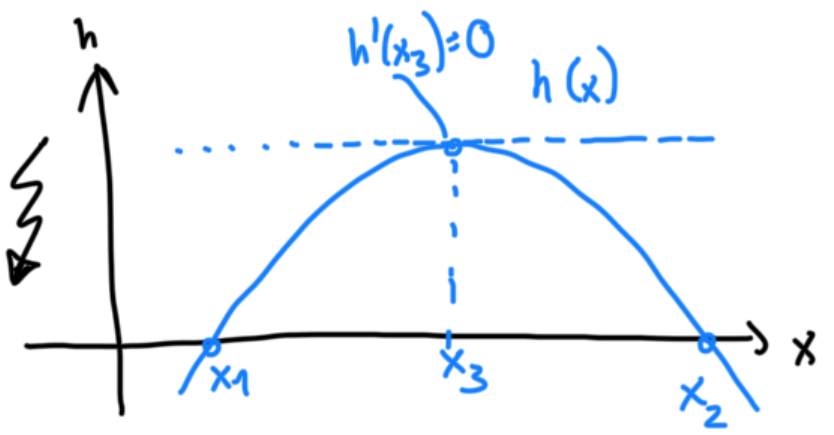
$h(x_2) = 0$, $x_1 < x_2$, $x_1, x_2 \in [a,b]$

Rolle's theorem (MVT) $\Rightarrow$ $\exists\, x_3 \in (a,b)$ s.t. $h'(x_3) = 0$

But $h'(x_3) = g'(x_3) - 1 \leq k - 1 < 0$

$-k \leq g'(x_3) \leq k$ (condition 3)

$k < 1$



contradiction!

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Example: $g(x) = \frac{x^2-1}{3}$ has a unique fixed point in $[-1,1]$.

Solution: Check conditions 1)–3)

1) g is continuous since it is a polynomial

2) g maps $[-1,1]$ to itself (graph it or use EVT)

- $g(-1) = 0$ ✓ (endpoints are ok)
- $g(1) = 0$ ✓
- $g'(x) = \frac{2}{3}x \overset{!}{=} 0 \Rightarrow x = 0$ $g(0) = -\frac{1}{3} \in [-1,1]$ ✓

only extremum in $(-1,1)$

3) $|g'(x)| = \left|\frac{2}{3}x\right| \leq \frac{2}{3} < 1$ for $x \in [-1,1]$ ✓

What is the fixed point? $g(p) = p = \frac{p^2-1}{3}$

$\Longleftrightarrow$ $p^2 - 3p - 1 = 0$

$p = \frac{3 \pm \sqrt{13}}{2}$

discard + sign value since not in $[-1,1]$

$$\Rightarrow \quad p = \frac{3-\sqrt{13}}{2} = \frac{-2}{3+\sqrt{13}}$$

$p = 3^{-p} - 2^{-p}$:)

Fixed point iteration: Pick $p_0 \in [a,b]$

Then set $p_n = g(p_{n-1}) \qquad n = 1, 2, \ldots$

$(= g(g(\ldots g(p_0))))$

Hope $g_n \to p = g(p)$

Fixed point theorem (Convergence of fixed point iteration)

Suppose g is continuous on $[a,b]$, maps $[a,b]$ onto itself, $\exists\ k<1$ such that $|g'(x)| \le k$ for $x \in [a,b]$. Then for any $p_0 \in [a,b]$ the sequence $p_n = g(p_{n-1})$, $n = 1, 2, 3, \ldots$ converges to p, the unique fixed point of g in $[a,b]$.

Proof:

$$|p_n - p| = |g(p_{n-1}) - g(p)| = |g'(\xi(p_{n-1}))(p_{n-1} - p)|$$

($\xi(p_{n-1}) \in [a,b]$)

($p_n = g(p_{n-1})$, $p = g(p)$)

(Taylor's theorem (*))

$$\le k\,|p_{n-1} - p| = (\Delta)$$

($|g'(\xi(p_{n-1}))| \le k$)

(*) $\quad g(p) = g(p_{n-1}) + g'(\xi(p_{n-1}))(p - p_{n-1})$

$$g(p_{n-1}) - g(p) = -g'(\xi(p_{n-1}))(p - p_{n-1})$$

$$(\Delta) \le k^2 |p_{n-2} - p| \le k^3 |p_{n-3} - p| \le \ldots \le k^n |p_0 - p|$$

same way

So we have $|p_n - p| \leq k^n |p_0 - p| \qquad n \geq 1$

$\downarrow$ $n \to \infty$ because $k < 1$

0

So $\lim_{n\to\infty} p_n = p$

$$p = \lim_{n\to\infty} p_n = \lim_{n\to\infty} g(p_{n-1}) = g\left(\lim_{n\to\infty} p_{n-1}\right) = g(p)$$

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Corollary 1: $|p_n - p| \leq k^n \max(p_0 - a, b - p_0)$

Corollary 2: $|p_n - p| \leq \frac{k^n}{1-k} |p_1 - p_0|$ p_0, p_1 are known

Proof: $|p_0 - p| \leq |p_0 - p_1| + |p_1 - p|$ (Δ-inequality)

$\leq |p_0 - p_1| + k|p_0 - p|$

$\Leftrightarrow (1-k)|p_0 - p| \leq |p_0 - p_1| \quad (0)$

$$|p_n - p| \leq k^n |p_0 - p| \overset{(0)}{\leq} \frac{k^n}{1-k} |p_0 - p_1|$$

$|p_n - p| \leq k^n |p_0 - p|$: Fixed point theorem

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Corollary 3: $|p_n - p| \leq \frac{k}{1-k} |p_n - p_{n-1}|$

Proof: $|p - p_n| \leq k |p - p_{n-1}| \qquad (*)$

$$|p_{n-1} - p| \leq |p_{n-1} - p_n| + |p_n - p|$$

$$\overset{\leq}{(\star)} \; |p_{n-1}-p_n| + k\,|p_{n-1}-p|$$

$$(1-k)\,|p_{n-1}-p| \leq |p_{n-1}-p_n| \qquad (\heartsuit)$$

$$\overset{(\star)\,\&\,(\heartsuit)}{|p-p_n| \leq \frac{k}{1-k}\,|p_{n-1}-p_n|}$$

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Could be used as a stopping criterion!

Remark: Could only require g to be a contraction on $[a,b]$

$$|g(x)-g(y)| \leq k|x-y| \quad \text{for} \quad k<1\,,\; x,y \in [a,b]$$

i.e., g is Lipschitz continuous with Lipschitz constant $k<1$.

Example: $g(x) = \frac{x^2-3}{3}$, $[a,b] = [-1,1]$, $p_0 = 0$

How many iterations do we need to achieve $|p_n - p| \leq 10^{-4}$?

We have $|g'(x)| = \left|\frac{2}{3}x\right| \leq \frac{2}{3}$ on $[-1,1]$

By Cor. 1. $|p-p_n| \leq \left(\frac{2}{3}\right)^n \max(\underbrace{p_0-a}_{1}, \underbrace{b-p_0}_{1}) = \left(\frac{2}{3}\right)^n \overset{\text{(want)}}{\leq} 10^{-4}$

So we need to solve $\left(\frac{2}{3}\right)^n \leq 10^{-4}$ for n:

$$n \underbrace{\log\left(\frac{2}{3}\right)}_{<0} \leq -4\log(10)$$

$$\Leftrightarrow \quad n \geqslant \frac{-4\log 10}{\log\left(\frac{2}{3}\right)} \cong 22.7 \Rightarrow n = 23 \text{ should be good enough}$$